Shivakanth Sujit

shivakanth.sujit@gmail.com | Tokyo, Japan | ShivaSujit | shivakanthsujit | shivakanthsujit

Profile

Role: Deep RL Researcher / Research Engineer

Location: Tokyo, Japan

Focus: Embodied AI, VLA/VLM post-training, reinforcement learning, and high-performance robotics deployment systems.

Experience

ARAYA Inc., Deep RL Researcher — Tokyo, Japan — 2024 – present — 2 years

- Building post-training and deployment frameworks for vision-language-action models used in assistive human-robot interaction.
- Demonstrated that a standard offline actor-critic objective can post-train a billion-scale flow-matching VLA without flow-specific actor objectives, distillation, or test-time guidance.
- Built batched inference and KV-cache support for a robotics foundation model, reducing a key rollout path from about 20 seconds to 0.2 seconds and enabling online RL.
- Optimized data, training, and inference pipelines by about 2x, enabling same-day data collection-to-deployment cycles.
- Built automated annotation pipelines that transform demonstrations and instructions into dense subtask and motion-primitive supervision.

Microsoft Research, Research Intern — New York, USA — 2023 – 2023 — 1 year

- Developed a reachability-aware latent representation using multi-step inverse dynamics for sample-efficient reinforcement learning.
- Enabled hierarchical planning with 3x faster goal reaching and 50% fewer operations, including zero-sample ad hoc goal reaching in reward-free environments.

Education

Mila / École de technologie supérieure, Computer Science — Montreal, Canada — 2021 – 2023

- GPA: 4.15/4.30
- Thesis: Evaluation of sample efficiency in offline reinforcement learning
- Supervisor: Prof. Samira Ebrahimi Kahou

National Institute of Technology Tiruchirappalli, Instrumentation and Control Engineering — 2017 – 2021

Research Highlights

Offline RL post-training for flow-matching VLAs (2026): Applied the vanilla critic-gradient actor objective to a billion-scale flow-matching VLA with behavior-cloning regularization; improved task success from a 4% behavior-cloning floor to 94% on a task-agnostic play-data benchmark; built chunked FQL, Cal-QL critics, exact 10-step BPTT with gradient checkpointing, and 24-arm sweeps across 8 GPUs.

Reducible Loss prioritization (2022–2023): Developed ReLo, a prioritization method that improved performance by 20% over prioritized experience replay with negligible overhead; published at NeurIPS 2023.

Variational Sparse Gating (2021–2022): Improved model-based RL sample efficiency with 2x fewer gradient steps; published at NeurIPS 2022.

Technical Strengths

Embodied AI and reinforcement learning: Offline and online RL; actor-critic methods; VLA/VLM post-training; behavior cloning; rollout evaluation; real-robot integration; assistive robotics; shared autonomy; and human-robot interaction.

Performance engineering: Profiling-driven GPU optimization, batching, KV-cache management, vectorization, kernel tuning, compilation, and end-to-end optimization across data collection, training, inference, and deployment.

Research systems: Reproducible experiment pipelines, large-scale sweeps, diagnostics, automated annotation, and deployment tooling.

Publications

Research output
Twelve papers and preprints, plus a master's thesis, spanning reinforcement learning, robotics, human-robot interaction, computer vision, and scientific machine learning. See the complete publication list or Google Scholar.
Shivakanth Sujit